Human-Computer Interaction (HCI)

Short English notes: model answers for Chapters 1 to 9

Chapter 1: The Human (insaan)

1.1 HCI mein 'Human' ko kyun samajhte hain?

The human is an information-processing system. Information comes in through the senses (vision, hearing, touch), is stored in memory (sensory, short-term, long-term), is processed by reasoning and problem solving, and results in actions. Emotion and individual differences also affect performance.

1.2 Vision (dekhna)

Vision has two stages: physical reception of the stimulus and processing/interpretation of it. The retina has rods (low light) and cones (colour). Colour has hue, intensity and saturation. Blue acuity is lowest, and about 8% of males and 1% of females are colour blind. Reading involves saccades and fixations, and perception happens during fixations.

1.3 Hearing (sunna)

Hearing provides information about distance, direction and objects. The ear has an outer, middle and inner part. Sound has pitch (frequency), loudness (amplitude) and timbre (quality). Humans hear about 20 Hz to 15 kHz, and can focus on one sound over background noise (the cocktail party phenomenon).

1.4 Touch (chhoona)

Touch gives feedback via skin receptors: thermoreceptors (heat/cold), nociceptors (pain) and mechanoreceptors (pressure). It is especially important for visually impaired users. Kinesthesis is the awareness of body position.

1.5 Movement aur Fitts' Law

Response time is reaction time plus movement time. Fitts' Law: Mt = a + b log2(D/S + 1), where Mt is movement time, D is distance, S is target size, and a and b are constants. Targets should be as large and as close as possible.

1.6 Memory (yaad dasht)

There are three memory types: sensory (iconic, echoic, haptic), short-term (limited to 7 ± 2 chunks, fast decay) and long-term (huge capacity, episodic and semantic). LTM models: semantic networks, frames, scripts and production rules. Rehearsal moves information to LTM; forgetting is due to decay and interference. Recognition is easier than recall.

1.7 Thinking (sochna): Reasoning

Deduction derives a logically necessary conclusion from premises. Induction generalizes from seen cases to unseen cases (it can only be disproved, never proved). Abduction reasons from an event to its likely cause and can give false explanations. Humans are poor at using negative evidence (Wason's cards).

1.8 Problem solving, Errors aur Mental models

Problem solving theories include Gestalt, problem space theory (states, operators, heuristics such as means-ends analysis) and analogy. Skilled users chunk information. A slip is a correct intention carried out wrongly; a mistake is a wrong intention caused by incorrect understanding. A wrong mental model of the system leads to errors.

1.9 Emotion (jazbaat)

Emotion involves both cognitive and physical responses. Affect is the biological response to stimuli: positive affect helps creative problem solving, negative affect narrows thinking. For design, stress makes problem solving harder, relaxed users forgive design flaws, and pleasing interfaces increase positive affect.

1.10 Individual differences aur Psychology ka istemal

People differ in long-term (abilities), short-term (stress, fatigue) and changing (age) ways. A designer should ask whether a design decision will exclude part of the user population. Psychology knowledge is distilled in guidelines, cognitive models and evaluation techniques.

Chapter 2: The Computer

2.1 Computer system ke hisse

A computer system consists of input devices, output devices, virtual reality devices, physical interaction (sound, haptic, bio-sensing), paper, memory and processing/networks. The devices dictate the styles of interaction a system can support.

2.2 Text entry devices

Text entry devices include keyboards (QWERTY, Dvorak, alphabetic, special, chord), phone pads with T9 predictive entry, handwriting recognition and speech recognition. The QWERTY layout is not optimal for typing but is kept because of the large social base of QWERTY typists.

2.3 Positioning, pointing aur drawing devices

Pointing devices include the mouse, touchpad, trackball, joystick, touch screen, stylus, digitizing tablet, eyegaze and cursor keys. A mouse is an indirect device because hand movement on the desk moves a cursor on the screen. A touch screen is a direct device: it is fast and needs no special pointer, but is imprecise and tiring for the arm.

2.4 Display devices (screens)

A bitmap display is made of many coloured pixels; resolution, aspect ratio and colour depth describe its quality. Anti-aliasing softens jagged edges. CRTs have health hazards (radiation, fields), while LCDs are lighter with no radiation problems. Other displays: large, situated (location-specific) and digital paper.

2.5 Virtual Reality aur 3D

Virtual reality uses 3D positioning devices (3D mouse with six degrees of freedom: x, y, z, roll, pitch, yaw), data gloves, helmets and body tracking. VR motion sickness comes from time delay (lag) and conflicting depth cues between eye angle and focus.

2.6 Physical controls, sensors aur sounds

Dedicated displays can be analogue or digital; head-up displays show key information in the user's line of sight. Sound gives feedback and warnings, and haptic devices give touch feedback (for example vibration, BMW iDrive). Sensors detect the environment and the user's body (RFID, heart rate, iris scan).

2.7 Paper: printing aur scanning

Printers form images from dots (dot-matrix, ink-jet, laser, thermal), measured in dpi and pages per minute. Fonts differ in size (points), pitch (fixed or variable) and serif or sans-serif. Page description languages such as PostScript describe a page instead of sending a bitmap. Scanners convert paper to bitmaps and OCR converts bitmaps back into text.

2.8 Memory aur storage

RAM is fast, volatile short-term memory; disks (magnetic, optical) are long-term storage; flash memory is persistent silicon memory. Virtual memory stores some programs temporarily on disk but swapping slows the system. Compression is lossless (ZIP, GIF) or lossy (JPEG, MP3).

2.9 Processing aur networks

Designers should not assume an infinitely fast machine. Moore's law says processor speed doubles roughly every 18 months. Interactive performance is limited by computation, storage, graphics and network capacity. Networks give access to large resources and other people but cause delays, conflicts and unpredictability.

Chapter 3: The Interaction

3.1 Interaction kya hai? Domain, Goal, Task

Interaction is communication between the user and the system. Domain is the area of work, a goal is what the user wants to achieve, and a task is how the goal is achieved in terms of actions.

3.2 Donald Norman ka Interaction Model (7 stages)

Norman's model has seven stages: establish the goal, formulate the intention, specify the action sequence, execute the action, perceive the system state, interpret it, and evaluate it against the goal. The gulf of execution is the difference between the user's intended actions and the actions the system allows. The gulf of evaluation is the difference between what the user expects and what the system shows.

3.3 Abowd aur Beale ka Interaction Framework

The interaction framework (Abowd and Beale) has four parts: User, Input, System and Output, each with its own language. Interaction is translation between these languages, and problems in interaction are problems in translation. The user's intention is translated into input actions, then into a change of system state, shown as output, and interpreted by the user.

3.4 Ergonomics aur Industrial interfaces

Ergonomics studies the physical characteristics of interaction (arrangement of controls and displays, environment, health issues, use of colour). Industrial interfaces differ from office interfaces: data is numeric, changes are fast and the environment is harsh. Glass interfaces replace dials with screens; industrial work often involves indirect manipulation of the real world.

3.5 Interaction styles (interface ke andaaz)

Common interaction styles: command line, menus, natural language, question/answer and query dialogue, form-fills and spreadsheets, WIMP, point-and-click and three-dimensional interfaces. Menus rely on recognition rather than recall, so they are easier for novices, while command lines suit experts.

3.6 WIMP ke elements

WIMP stands for windows, icons, menus and pointers. Windows are independent areas of the screen, icons are small pictures representing objects or actions, menus offer choices, and pointers select items. Related elements are buttons (radio buttons for exclusive choices, check boxes for non-exclusive), toolbars, palettes and dialogue boxes.

3.7 Interactivity: look, feel, initiative, error

Look and feel is appearance plus behaviour; different window systems with the same elements behave differently. In WIMP systems the user has the initiative, except for modal dialogue boxes. Errors cannot always be avoided, so systems should make errors easy to detect and repair. Social and organizational context also affects interaction.

3.8 Experience, engagement aur fun

It is not enough that people can use a system; they must want to use it. Flow is the balance between anxiety and boredom. Designers must consider experience elements such as excitement, suspense, surprise and shared experience in addition to function.

3.9 Physical design aur Value

Physical design involves constraints (ergonomic, physical, legal and safety, context, aesthetic, economic) that often conflict, so trade-offs are needed. People use something only if its perceived value exceeds its cost. Because people heavily discount future value, designs need low barriers and high perceived present value.

Chapter 4: Paradigms (HCI ki tareekh)

4.1 Paradigm kya hota hai?

A paradigm is a predominant theoretical framework or world view. In HCI, a paradigm shift occurs when a new computing technology creates a new perception of the human-computer relationship. Studying HCI history means studying a series of such shifts.

4.2 Paradigm shifts ka silsila

Key HCI paradigms: batch processing, time-sharing, video display units (Sketchpad), programming toolkits, personal computing, window systems and WIMP, metaphor, direct manipulation (Shneiderman), hypertext, multimodality, CSCW, the World Wide Web, agent-based interfaces, ubiquitous computing (Weiser) and sensor-based, context-aware interaction. Direct manipulation means visibility of objects, incremental action with rapid feedback, reversibility, and replacing language with action.

Chapter 5: Interaction Design Basics

5.1 Design kya hai?

Design is achieving goals within constraints. The golden rule of design is to understand your materials: computers and people. Human error is normal, so systems should be designed for it; users should be treated at least as well as physical materials.

5.2 Design ka process

The design process includes requirements, analysis, design, iteration and prototyping, and implementation and deployment. Because time is limited, designers must make trade-offs about which usability problems to find and fix.

5.3 User focus: Persona aur Cultural probes

A persona is a description of an example user, not necessarily a real person, used as a surrogate user. Details make the persona feel real. Cultural probes are probe packs (for example camera, postcard) given to people so they record what is meaningful to them in their own environment.

5.4 Scenarios (design ki kahaniyan)

A scenario is a story for design: a step-by-step walkthrough of what users do, see and think. Scenarios help communicate with others, validate other models and express dynamics. Their weakness is linearity: they show one path and miss alternatives, so several scenarios and methods should be used.

5.5 Navigation design

Navigation design works at local (single screen), global (whole site or application) and wider (other applications) levels. The four golden rules of navigation: know where you are, know what you can do, know where you are going or what will happen, and know where you have been or what you have done. Breadcrumbs show the path through a hierarchy.

5.6 Screen design aur layout

Screen design relies on grouping and structure (logically related items physically together), order that matches the task, decoration, alignment (left-align text, align decimal points for numbers) and white space to separate, structure and highlight items.

5.7 User action aur control: forms aur affordance

Affordance is a psychological term for physical objects: the shape and size of an object suggest the actions it allows, for example a mug handle affords grasping. On screen, a button-like object affords clicking. Forms need logical grouping, natural order and a correct tab order.

5.8 Information present karna: aesthetics, colour, culture

Aesthetically pleasing designs increase user satisfaction, but beauty and utility may conflict. Colour and 3D are often misused: use colour sparingly to reinforce other information and consider colour blindness. Localisation changes interfaces for particular cultures and languages; symbols may have different meanings in different cultures.

5.9 Iteration aur prototyping

Iteration and prototyping: you never get it right the first time, so design, prototype, evaluate and re-design until the result is acceptable. Pitfalls: a good starting point is needed and the designer must understand what is actually wrong, or the design may only reach a local improvement.

Chapter 6: HCI in the Software Process

6.1 Software lifecycle

The waterfall model has requirements specification, architectural design, detailed design, coding and unit testing, integration and testing, and operation and maintenance. Verification means designing the product right; validation means designing the right product. For interactive systems the lifecycle cannot be linear because there is a lot of feedback between stages.

6.2 Usability engineering

Usability engineering makes specific usability measures explicit as requirements. A usability specification gives the attribute, measuring concept, measuring method and the now, worst-case, planned and best-case levels. ISO 9241 defines usability as effectiveness, efficiency and satisfaction.

6.3 Iterative design aur prototyping

Prototypes simulate or animate features of the intended system; types are throw-away, incremental and evolutionary. Techniques include storyboards, limited-functionality simulations and the Wizard of Oz technique. A risk of iterative design is design inertia: early bad decisions stay bad.

6.4 Design rationale

Design rationale is information that explains why a system is the way it is. Benefits: communication across the life cycle, reuse of design knowledge, design discipline, recording of trade-offs. Process-oriented rationale (IBIS: issues, positions, arguments) keeps the order of decisions; structure-oriented rationale (QOC: questions, options, criteria) structures the considered alternatives.

Chapter 7: Design Rules

7.1 Design rules ki qismein

Design rules are principles (abstract, low authority, high generality), guidelines (more specific, lower authority, wider application) and standards (specific, high authority, limited application). Standards need a sound underlying theory and slowly changing technology; ISO 9241 defines usability as effectiveness, efficiency and satisfaction.

7.2 Usability ke principles: Learnability, Flexibility, Robustness

Principles to support usability: learnability (predictability, synthesizability, familiarity, generalizability, consistency), flexibility (dialogue initiative, multithreading, task migratability, substitutivity, customizability) and robustness (observability, recoverability, responsiveness, task conformance).

7.3 Golden rules aur heuristics

Shneiderman's eight golden rules: strive for consistency, enable shortcuts for frequent users, offer informative feedback, design dialogs to yield closure, offer error prevention and simple error handling, permit easy reversal of actions, support internal locus of control, and reduce short-term memory load.

7.4 HCI design patterns

A design pattern is an invariant solution to a recurrent problem within a specific context. Patterns capture design practice, represent knowledge at varying levels, are easy to read for all stakeholders and, linked together in a pattern language, help generate complete designs. Example: 'Go back to a safe place'.

Chapter 8: Implementation Support

8.1 Programming tools ke levels

Programming support has three layers: windowing systems (core support for separate and simultaneous user-system activity), interaction toolkits (bring programming closer to the user's perception) and user interface management systems (control the relationship between presentation and functionality).

8.2 Windowing systems

A windowing system provides device independence (an abstract terminal and image models such as pixels or PostScript) and resource sharing among independent processes. Three architectures exist: each application manages everything, management inside the OS kernel, or management as a separate application (most portable). X Windows uses a client-server architecture with a separate window manager.

8.3 Application programming: event loop aur notification

In the read-evaluation (event) loop style, the program loops reading events and handling them by type. In the notification-based style, the programmer registers callbacks for events and the system calls them. A modal dialog is easy with an event loop and hard with notification; a non-modal dialog is the opposite. Implementation should not drive design.

8.4 Toolkits

Toolkits provide interaction objects (widgets) that link input and output, promoting consistency and generalizability through a similar look and feel. Java's AWT is notification-based and Swing is built on AWT using the MVC architecture.

8.5 UIMS (User Interface Management Systems)

A UIMS adds a level above toolkits. As a conceptual architecture it separates application semantics from presentation, which improves portability, reusability, support for multiple interfaces and customizability. Examples of UI development environments include Visual Basic.

8.6 Seeheim model aur Arch/Slinky

The Seeheim model divides an interface into presentation (lexical), dialogue control (syntactic) and functionality (semantic), with a switch/bypass for rapid semantic feedback. Arch/Slinky adds more layers and distinguishes lexical from physical; layers may be thicker or thinner in different systems.

8.7 MVC aur PAC

MVC (Model-View-Controller) separates the internal logical state (model), how it is rendered (view) and input processing (controller). In practice the controller talks to the view, so the separation is incomplete. PAC (Presentation-Abstraction-Control) is closer to Seeheim: abstraction holds the logical state, presentation handles input/output and control mediates between them, managing hierarchy and multiple views.

8.8 Dialogue controller ke techniques aur graphical specification

Dialogue controller techniques include menu networks, state transition diagrams, grammar notations, event languages, declarative languages, constraints and graphical specification. Graphical specification (Visual Basic, Dreamweaver, Flash) lets the designer draw components and attach actions, but it is hard to see the paths through the whole system.

Chapter 9: Evaluation Techniques

9.1 Evaluation kya hai?

Evaluation tests the usability and functionality of a system and should be done at all stages of the design life cycle. Its goals are to assess the extent of system functionality, assess the effect of the interface on the user, and identify specific problems.

9.2 Designs ko evaluate karna (users ke baghair)

Cognitive walkthrough (Polson et al.) evaluates how well a design supports the user in learning a task, by an expert walking through it. Heuristic evaluation (Nielsen and Molich) has experts check whether a design violates agreed usability heuristics. Review-based evaluation uses results from the literature, and model-based evaluation uses cognitive models such as GOMS.

9.3 Users ke saath evaluation: Laboratory vs Field

Laboratory studies offer specialist equipment and an uninterrupted environment but lack context. Field studies take place in the natural environment, keep context and allow longitudinal studies, but suffer distractions and noise.

9.4 Experimental evaluation

Experimental evaluation tests a hypothesis by comparing conditions that differ only in a controlled variable. The independent variable is what is changed; the dependent variable is what is measured. The null hypothesis states there is no difference between conditions and the aim is to disprove it. In a within-groups design each subject does every condition; in a between-groups design each subject does only one.

9.5 Observational methods

Observational methods: think aloud (the user describes what they are doing and why), cooperative evaluation (user and evaluator collaborate and can ask each other questions), protocol analysis (paper, audio, video, computer logging, user notebooks) and post-task walkthroughs (the user comments on a recording after the task).

9.6 Query techniques aur Physiological methods

Query techniques: interviews (one-to-one, flexible but subjective and time consuming) and questionnaires (fixed questions, quick and reach many users but less flexible). Eye tracking measures fixations, saccades and scan paths. Physiological measures include heart activity, galvanic skin response (GSR), EMG and EEG, but they are difficult to interpret.

9.7 Evaluation method kaise chunein?

Choosing an evaluation method depends on when in the process (design or implementation), laboratory or field, how objective, qualitative or quantitative measures, level of information, level of interference, and the available resources (time, subjects, equipment, expertise).